

Ancient Script Decipherment

Phase 1: Data, Census, and Frozen Splits

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1 Summary

Eight corpora ingested with provenance and content hashes; 19 test partitions frozen (SHA-256 manifest committed to git) and *enforced* by the access guard; census committed and regenerable. No model training occurred: every step in this phase is download plus file processing (heaviest single step: Coptic deduplication, 15 s of CPU). **make check** is green (lint, 80 tests, no-pretrained gate, smoke).

2 Census

corpus	sentences	docs	tokens	translation	source
tla_earlier_egyptian	12,773	95 ¹	70,267	German	TLA (HF, CC-BY-SA-4.0)
tla_late_egyptian	3,606	34 ¹	24,437	German	TLA (HF, CC-BY-SA-4.0)
tla_demotic	13,383	130 ¹	117,314	German	TLA (HF, CC-BY-SA-4.0)
coptic_scriptorium	52,105	1,490	736,585	En (partial)	C. SCRIPTORIUM
hebrew_morphhb	23,213	39	306,785	—	OpenScriptures WLC
ugaritic_hebrew_cog.	43,951 pairs	—	—	—	NeuroDecipher
linearb_greek_cog.	919 pairs	—	—	—	NeuroDecipher
logogram_nlp	inventory: LNA	165 MB, EGY 10 MB, ZHO	138 MB	—	LogogramNLP

Table 1: ¹Dating cohorts, not true documents (§3). Full machine-readable census: `docs/census/census.{md,json}`. Both decipherment-ladder rungs’ cognate data come from Luo et al.’s own release — rung 2 (Linear B→Greek) is runnable without DĀMOS.

3 TLA pseudo-documents (the honest caveat)

The official TLA exports carry no text IDs and are not text-ordered (contiguous same-date runs $\approx$ 1 sentence). `doc_id` is therefore the **dating cohort**: all sentences sharing the exact (dateNotBefore, dateNotAfter, authors) triple. A real text’s sentences always share these values, so no text can straddle partitions; cohorts over-merge (95/34/130 groups), which errs strict, never leaky. Obtaining real text IDs from the TLA maintainers is an open gap that would upgrade cohort-held-out to true document-held-out.

4 Splits and freezing

Seed 1234, ratios 80/10/10 by record weight over whole groups; every split ships `split_info.json`; all 19 test partitions are SHA-256-frozen in the committed manifest. The guard now *enforces*

freezing: reading a frozen test file re-hashes it (tamper $\Rightarrow$ hard error) and write-mode opens of frozen files are refused — tested in both directions.

Headline scheme = **dedup**: cohort/document-held-out plus cross-split near-duplicate removal (normalized edit distance ≤ 0.1 ; exact-hash plus banded Levenshtein behind a bag-of-characters lower bound).

corpus	removed (exact + near)	test after dedup
tla_earlier_egyptian	857 + 107	718
tla_late_egyptian	54 + 8	662
tla_demotic	171 + 54	1,283
hebrew_morphhb	50 + 9	3,227
coptic_scriptorium	1,597 + 249	4,541

Table 2: Dedup removals (always reported, never silent). Earlier Egyptian loses $\sim 34\%$ of naive test/valid to near-verbatim train overlap — precisely the contamination mode that inflated published hieroglyph-MT results by 29–47 BLEU.

Also materialized: **sign_heldout** (25 mid-frequency sign types held out entirely; 197/335 test sentences for Earlier/Late), and **period_heldout** (century buckets; Earlier + Demotic — Late Egyptian has only 8 buckets and the scheme is dropped there with a note). **site_heldout**: no Phase 1 corpus carries site metadata. **random** exists as a labeled diagnostic only.

5 Environment note

The Desktop/iCloud problem from Phase 0 *recurred*: overnight, 6,760 venv files were re-flagged macOS-hidden, which makes Python ≥ 3.12 skip `.pth` files and silently breaks imports. Durable fix: `.venv` is now a symlink to `~/venvs/glyphos` (outside iCloud), matching `data` $\rightarrow$ `~/dev/glyphos-data`. No mutable tool state remains inside iCloud.

6 Next: Phase 2 — rendering engine

PangoCairo + Noto (incl. Egyptian Hieroglyphs), 24×24 windows with stride 12, the composable degradation suite with **damage_level**, the multiscrypt self-supervised stream, and `<g>...</g>` sign-markup handling. Requires cairo/pango system libraries; install path for this machine decided at phase start.

Addendum (second pass, 2026-08-02): frontier downloads

Every remaining openly-obtainable corpus was downloaded and integrated — the census now counts **11 ingested corpora and 26 frozen test partitions**: First1KGreek (392,478 passages, 28.8M tokens; the cluster deferral was reversed — parsing cost 15 s), CDLI cuneiform (132,210 tablets with catalogue language/period; categorical period-held-out), and the first machine-readable **Meroitic** corpus (Otten & Anastasopoulos 2025; 18,103 lines; its bundled pretrained embeddings are quarantined under the sealed constraint). Mayan and Libyco-Berber have no open bulk source (MHD application-gated; the LBI database is defunct and its domain squatted — recoverable from pre-2020 Wayback captures); concrete unblock paths are filed in `docs/data_gaps.md`. Dedup for the

two large monolingual LM corpora is exact-duplicate-only (Greek -617 , CDLI $-1,891$); TLA headline splits keep the full near-duplicate pass.